

The $\cos 2\phi$ modulation on tensor-polarized ^6Li : projected inclusive and coherent measurements

2026-08-17 (development run 5) · revised 2026-08-25 · polligen (evgen/) + polli_fastsim (fastsim/) · companions: plans/06 (background budget), docs/note_cos2phi_coherent_6Li.md, reports/polarized_li_primer.html (pedagogical introduction), reports/reconstruction_chain_report.html (the reconstructed-level counterpart: money plots 5R, 7R, 6R) · external numbers verified against primary sources · modification history in Appendix A

This note presents two projected EIC measurements of the $\cos 2\phi$ azimuthal modulation on a transversely tensor-polarized ^6Li beam: inclusive DIS, requiring only the scattered electron, and coherent diffraction with the ^6Li detected intact in its 1^+ ground state. The modulation arises from photon double helicity flip and probes gluon transversity, to which neither quarks nor bound nucleons contribute at leading twist [1,2]; it has not been measured on any target. The projections are given together with the assumptions entering each number and the checks performed on them.

13.5%

coherent-tag acceptance at IP6 with the 0.20 GeV envelope cut, $\exp(-B \cdot p_T^2 / \text{cut})$, $B = 50 \text{ GeV}^{-2}$ (9–20% for $B \in 40\text{--}60$). The cut is documented for top-rigidity beams; for ^6Li it is an optics assumption, and as an angular envelope it gives 9% at 50 GeV/u (§4.1).

Measured 2026-08-26: an intact ^6Li shot through the ePIC far-forward geometry clears the pots only above $|\theta_x| \approx 1.35 \text{ mrad}$ at this configuration — a tighter cut than either envelope, and open horizontally rather than vertically — which takes the tagged fraction to 2.7×10^{-3} and flips the sign of the acceptance-induced ($\cos 2\beta$) (tools/fullsim, plans/04 #20). Re-deriving the coherent projection on the measured aperture is open work.

1.1×10^7 / 1.1×10^8

tagged coherent events in 1 / 10 EIC years (scenario $f_0 = 0.04$, constant 0.20 GeV cut); best-bin $\delta A = 1.9 \times 10^{-3}$ / 6×10^{-4} , i.e. 5σ floors at 1.0% / 0.3% modulation. Reconstructed level (angular cut, slot cutout): 2.7×10^6 in year 1 (§4.1).

0.036

deformation-anchored (a_2) of the tagged sample at $P_{zz} = 0.6$ (band 0.018–0.059; one-sided $\times 0.73$ rate-weighting systematic), scaled from the polarized-deuteron CGC calculation [13]; sign flip vs the deuteron predicted

1 · The observable

For unpolarized electrons on a spin-1 target whose spin axis makes angle θ_m with the beam, the inclusive cross section per spin projection m is [1,2]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[F_1 + \frac{2}{3} a_m b_1 + \frac{1-y}{xy^2} \left(F_2 + \frac{2}{3} a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_m \Delta(x, Q^2) \cos 2\phi \right]$$

$$a_m = \frac{1}{4} c_m (3 \cos^2 \theta_m - 1), \quad c_m = 3 |\lambda_m| - 2 \rightarrow (1, -2, 1)$$

Sign convention of the tensor-rate terms. The b_1, b_2 terms above are written in the transcription used throughout this program. In the Hoodbhoy–Jaffe–Manohar convention as applied by HERMES and by Cosyn et al. (W. Cosyn, B. Roldan Tomei, A. Sosa, A. Zec, Eur. Phys. J. A 61 (2025) 83, Eq. 27: $A_{zz} = -(2/3) b_1 / F_1$ for the axis along the photon) the tensor-rate terms carry the opposite sign — the $m = 0$ state has the larger rate for $b_1 > 0$. This affects only the sign of A_{zz} relative to b_1 (and of the constant term of the spin-state ratio), not its magnitude, its error, or anything in the Δ sector; whether to flip the transcription or to document $b_1 \equiv -b_1^{\text{HJM}}$ is an open decision (docs/code_review_2026-08-25.md, item G1).

The $\cos 2\phi$ term requires the virtual photon to flip its helicity by two units. Since no spin- $1/2$ constituent can absorb two units of helicity, neither quarks nor bound nucleons contribute at leading twist: $\Delta(x, Q^2)$ is purely gluonic and vanishes for a nucleus composed of nucleons alone [1]. At leading order it is generated by the gluon transversity distribution δG through the $O(\alpha_s)$ photon–gluon convolution

$$\Delta(x, Q^2) = -\frac{\alpha_s(Q^2)}{2\pi} \left(\sum_q e_q^2 \right) x^2 \int_x^1 \frac{dy}{y^3} \delta G(y, Q^2)$$

(in the sign convention of Ref. [25]; conventions differ between papers). Δ has not been measured for any target; the only prior experimental document is a JLab letter of intent without a sensitivity projection [3]. Quantitative expectations are scarce. The only moment estimate is the Sather–Schmidt bag-model calculation for the spin-3/2 Δ^{++} baryon, $\int x \Delta dx = -0.012 \alpha_s$ [4], which suggests asymmetries at the per-mille level; percent-level values arise only in maximal-transversity ($\Delta_{TG} = \Delta_g$) scenarios for proton–deuteron Drell–Yan, described by their authors as optimistic [5]. The projections therefore span the scenario band $\Delta/F_1 \in 10^{-3}$ – 10^{-2} .

The measured amplitude and its inversion back to the structure function are

$$A^{\cos 2\phi} = -\frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2, \quad \hat{\Delta} = -\hat{A} \frac{y^2 D_\phi}{1-y}$$

The fastsim approximation set is $A_{\parallel} = D \cdot g_1 / F_1$ (the y^2 -suppressed term dropped), $g_2 = g_2^{WW}$, and no target-mass corrections. The last matters only at high x and low Q^2 ; the sweet-spot bins sit at $x \leq 0.14$, where $y^2 = 4M^2 x^2 / Q^2 \leq 0.023$ (below 10^{-2} for $x \leq 0.06$), so it is a labeling caveat rather than a bias. One finite- y effect is not a labeling matter: the tensor-polarized b_1 – b_3 terms feed the $\cos 2\phi$ amplitude at $O(y^2)$, $\Delta_{\text{fake}} \approx y^2 b_1 / 6$ — at most 3% of the Δ amplitude at the sweet spots, but tens of percent of a 10^{-3} scenario at $x \approx 0.2$, $Q^2 \approx 1 \text{ GeV}^2$, where the amplitude-versus- x panels extend; it is computable from the same data's A_{zz} (reconstruction-chain report §2). Structure functions are toy parameterizations with PDF-grid (CT18NLO/NNPDFpol11) backends behind the same interface: rates are reliable to a factor of ≈ 1.5 , and asymmetry shapes are scenarios by construction.

2 • Machinery, statistics, and binning

Generator. No public generator produces polarized $e+A$ events for $A > 1$ (BeAGLE is unpolarized; DJANGO and PEPSI are nucleon-level and longitudinal-only). polligen therefore implements only the missing component, the spin-density matrix $\otimes$ structure-function master formula with spin-correlated sampling, and relies on existing tools for the rest. Its validation gates are bin-wise identity with the analytic asymmetry formulas at relative tolerance 10^{-12} , pseudo-experiment spreads that close on every analytic error map, and unbiased $\cos 2\phi$ recovery under ϕ -acceptance holes.

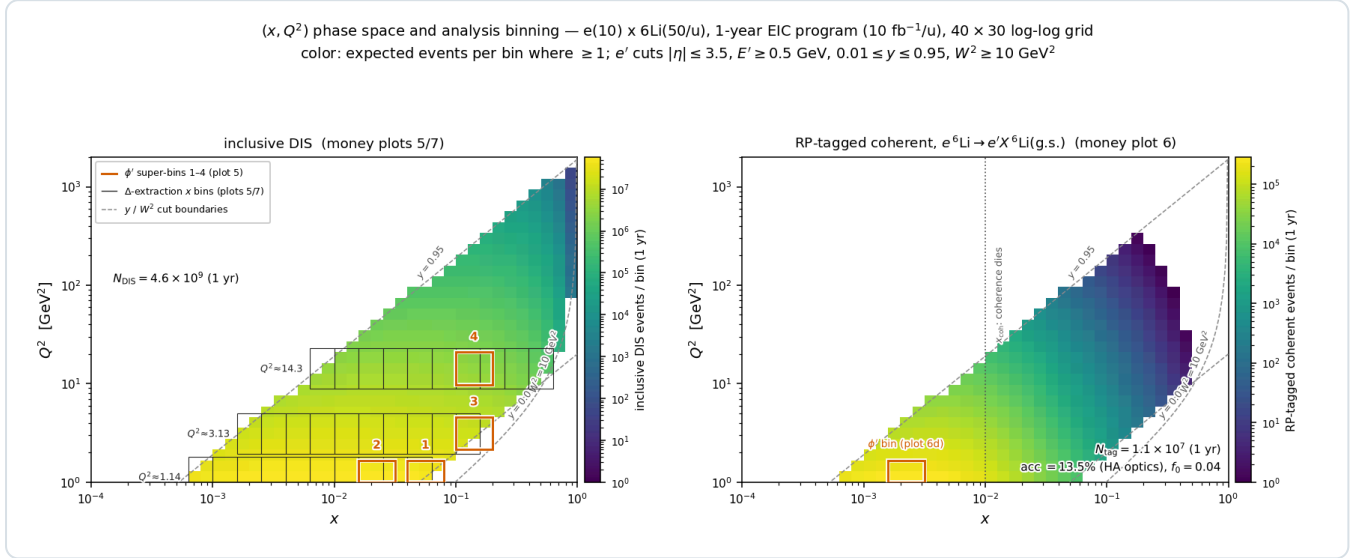
Full-luminosity statistics. For a binned estimator, drawing each ϕ -bin count from a Poisson distribution with the exact expected yield is statistically identical to event-level sampling, so a full-luminosity pseudo-experiment costs 24 Poisson draws (`sample.phi_histogram_pseudo`). Event-level sampling remains available and is what the closure tests use.

Estimator of the figures. The amplitudes shown below come from a single spin state: a binned least-squares fit of $N(\phi') = C(1 + A \cos 2\phi')$, not the $2\langle \cos 2\phi' \rangle$ moment (with realistic ϕ holes the moment is biased by more than the signal, while the fit stays unbiased), with $\delta A = \sqrt{(2/N)/P_{zz}}$ validated by pseudo-experiments and no $\cos \phi'$ term, since the m -symmetric tensor fills set $a_1 = 0$ by construction.

Estimator of the measurement. A single-state fit measures, besides the physics, the detector's own $\cos 2\phi'$ acceptance harmonic divided by P_{zz} : a 3% harmonic aligned with the vertical spin axis fakes an amplitude four times the signal. The estimator of the measurement is therefore the spin-state-sorted ratio of $m = \pm 1$ -rich ($P_{zz} = +0.6$) and $m = 0$ -rich (-1.2) bunches (reconstruction-chain report §3), which cancels any acceptance common to both states bin by bin, turns relative-luminosity errors into a constant, and has $\delta \hat{A} = 2\sqrt{(2/N)/(P_+ - P_0)} = 0.67\times$ the single-state error at the same source purity. At the reconstructed level — EMCal energy, track angles, $\epsilon_{\text{ID}}(\eta)$, hadronic-method y with the documented 25% resolution, reconstructed (x, Q^2) bins — the four sweet spots keep purities of 0.64–0.68 and give $\delta \hat{A} = 1.2 / 1.0 / 1.8 / 3.0 \times 10^{-4}$ in year 1 (money plot 5R), 0.64–0.70 of the values annotated in money plot 5; the extracted Δ (7R) reaches $\delta \Delta = 1.0 \times 10^{-3}$ at $Q^2 = 1.14 \text{ GeV}^2$ and 0.5×10^{-3} at 3.13 GeV^2 in year 1. Two caveats travel with those numbers: the bin-centering factor K is model-dependent at the 3–11% level between the repository's Δ shapes, larger than the 10-year statistical errors, and the 25% resolution is an absolute 50–125 MeV on a hadronic $E - p_z$ sum of 0.2–0.5 GeV at spots 1–3, a regime no published ePIC study covers. The cancellation presumes that both spin states see the same acceptance: the states must alternate bunch by bunch, since a 10^{-3} fill-to-fill difference of the $\cos 2\phi'$ efficiency harmonic fakes half the signal. The

figures below keep the single-state, truth-binned projections; their error bars are $1.4\text{--}1.6\times$ those of the reconstructed-level analysis.

Binning. The four displayed (x, Q^2) super-bins are selected by statistical significance within Q^2 bands: an unconstrained ranking concentrates at the lowest Q^2 , while the Q^2 lever arm provides the consistency check against power-suppressed contributions (§6). Each super-bin merges 3×3 cells of the 40×30 analysis grid. Rates use per-nucleon luminosity, F_2^A/A , scattered-electron cuts $|\eta| \leq 3.5$, $E' \geq 0.5$ GeV, $0.01 \leq y \leq 0.95$, $W^2 \geq 10$ GeV², and the full luminosity in the transverse-tensor fill; projections are quoted for the 1-year ($10 \text{ fb}^{-1}/u$) and 10-year ($100 \text{ fb}^{-1}/u$) programs.



Phase-space coverage and analysis binning (e10 × ⁶Li50, 1-year program, 40×30 log-log grid; color: expected events per bin where ≥ 1; dashed lines: y and W² cut boundaries). **Left, inclusive DIS** (4.6×10⁹ accepted events in year 1): boxes 1–4 are the ϕ' super-bins of money plot 5; gray combs are the three Q^2 slices ($Q^2 \times [1/1.6, 1.6]$) whose merged x-bin pairs form the data points of the amplitude-versus-x panel and of the Δ extraction (plot 5, right panel, and plot 7), retained where $N \geq 10^3$ and $\delta A \leq 8 \times 10^{-3}$. **Right, RP-tagged coherent channel:** the same map multiplied by $f_{\text{coh}}(x)$ and the 13.5% tag acceptance. $N_{\text{tag}} = 1.1 \times 10^7$ in year 1; 98% of the tagged sample lies at $x < 0.03$ (median 3.5×10^{-3}). The box is the super-bin of money plot 6(d); plot 6(c) is this map summed over Q^2 .

3 • The Δ model and the moment constraint

Provenance. Sather and Schmidt computed the lowest moment of Δ in the MIT bag model for the spin-3/2 Δ^{++} baryon, together with its Q^2 evolution; they provide no x dependence [4]. The explicit $\alpha_s(Q^2)$ is the leading-order Wilson coefficient of the photon–gluon box, so the $O(\alpha_s)$ suppression relative to F_1 is model-independent; the value -0.012 is the bag estimate. Lattice QCD provides two related results, each with substantial caveats: the bare ϕ -meson matrix element $A_2 = 0.23(2)(5)$ at $m_\pi = 450$ MeV [25] converts to $\int x \Delta|_\phi \approx -0.006 \alpha_s$ (our conversion, unrenormalized), while the NPLQCD deuteron calculation gives a δG moment of $-0.010(3)$, bare at $m_\pi \approx 806$ MeV [27], a transversity-to-momentum ratio roughly ten times below that of the ϕ and implying the opposite sign for the deuteron's Δ moment. No calculation of Δ exists for ⁶Li or any $A > 2$ nucleus; the only nuclear model is a $\Delta\Delta$ -isobar estimate for the deuteron [34]. The literature thus brackets the nuclear Δ in both directions: suppressed, as in the NPLQCD deuteron, or enhanced by binding in analogy with the EMC effect, as argued in the polarized-ion white paper [50].

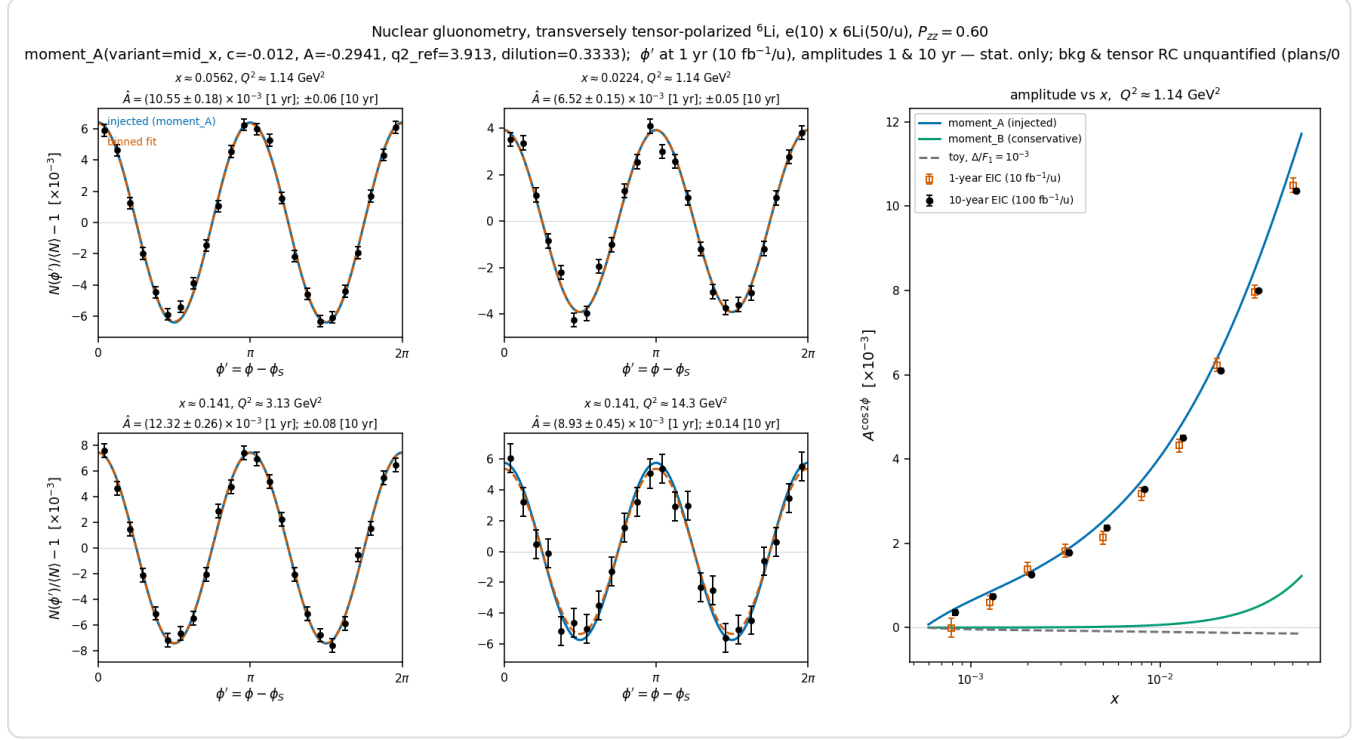
Implementation. All Δ models reside in one registry (`polli_fastsim/delta_models.py`). The production default, `moment_A`, is an ansatz normalized to the bag moment applied to the per-nucleon Δ of ⁶Li:

$$\Delta_A(x, Q^2) = \frac{1}{3} A \alpha_s(Q^2) F_1(x, Q^2) \frac{x^a(1-x)^b}{N_{\text{peak}}}, \quad \int_0^1 x \Delta(x, \langle Q^2 \rangle) dx = -0.012 \alpha_s \Rightarrow A = -0.294$$

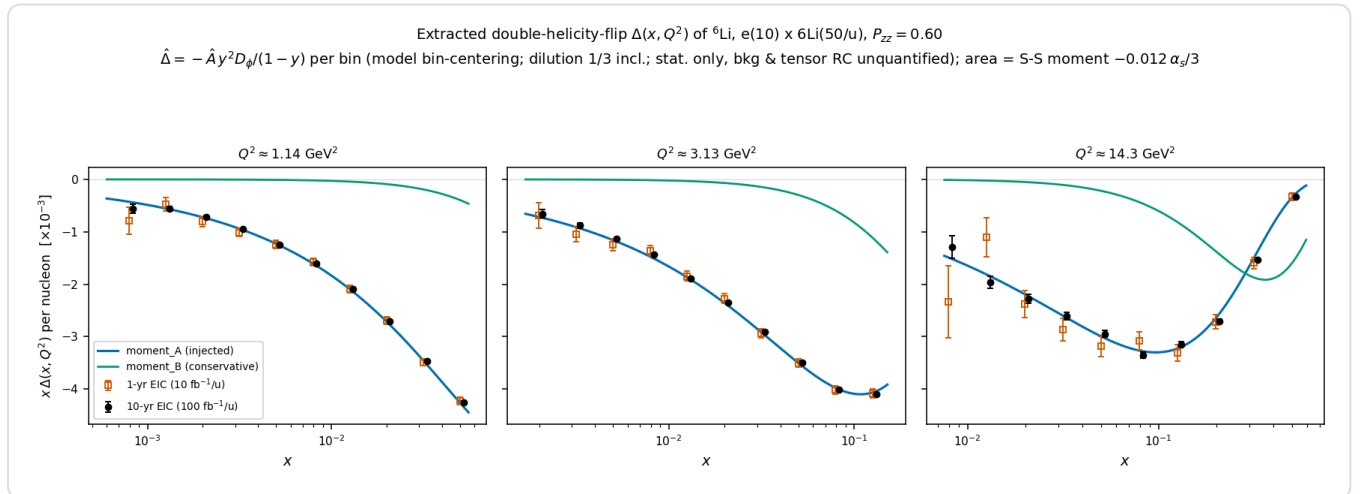
with (a, b) = (0.7, 3); the factor 1/3 restricts the exotic glue to the polarized deuteron-like pair, while the whole-nucleus P_{zz} remains a separate factor, so that no double counting occurs. The constraint is imposed at the rate-weighted $\langle Q^2 \rangle = 3.9$ GeV² of the accepted phase space; because F_1 evolves, the realized moment varies between -4% and $+7\%$ across the plotted Q^2 slices. α_s is taken from the CT18NLO table (the leading-order analytic fallback is 14% higher). The solved $A =$

-0.294 is consistent with the EPPS21-based money_delta value of -0.31 at its $\langle Q^2 \rangle = 7.4 \text{ GeV}^2$. **moment_B** applies the same constraint without the F_1 factor and yields amplitudes a factor of 2–8 lower; the A-versus-B spread is the dominant ansatz systematic, and both curves appear on every plot.

Status of the normalization. Transferring a single-hadron bag moment to a per-nucleon nuclear structure function has no precedent in the literature; both the normalization and the x dependence are labeled scenario choices, bracketed by **moment_B** from below and by the possible nuclear enhancement from above. The projections demonstrate that the measurement has the precision to decide: under the moment constraint, the x dependence separates interpretations A and B bin by bin.



Money plot 5 — inclusive gluonometry ($e10 \times {}^6\text{Li}50$, $P_{zz} = 0.60$, Δ : moment_A, dilution 1/3). Left 2x2: ϕ' pseudo-data with 1-year statistical error bars in the four sweet-spot super-bins ($Q^2 = 1.1 \rightarrow 14 \text{ GeV}^2$, $x \approx 0.02$ –0.14); injected curve (blue), binned fit (vermillion); annotations give δA for both programs. Right: extracted amplitude vs x with 1-year (open) and 10-year (filled) error bars against the moment_A, moment_B, and toy- 10^{-3} curves. Amplitudes are $(0.7$ – $1.2) \times 10^{-2}$ at the sweet spots against $\delta A = (1.5$ – $4.5) \times 10^{-4}$ in year 1, i.e. every bin $\geq 21\sigma$, so the A-versus-B spread is resolved by the x dependence. Only the scattered electron is detected.



Money plot 7 — the extracted $\Delta(x, Q^2)$. The amplitudes of money plot 5 unfolded to the structure function (model bin-centering, $\delta \Delta = \delta A \cdot |\partial \Delta / \partial A|$), shown as $x \Delta$ per nucleon at the three sweet-spot Q^2 slices with independent 1-year (open) and 10-year (filled) pseudo-measurements against the moment_A and moment_B curves; the area under each curve is the ported bag moment (± 3 dilution). Best bins reach $\delta \Delta \approx (0.7$ – $1.4) \times 10^{-3}$ in year 1 on $\Delta \approx -0.03$... -0.09 .

4 · The coherent channel

4.1 · Detecting the intact ${}^6\text{Li}$ at ePIC

In coherent diffraction, $e\ {}^6\text{Li} \rightarrow e'\ X\ {}^6\text{Li}(\text{g.s.})$, the recoil retains the beam's magnetic rigidity ($A/Z = 2$; the diffractive momentum loss $x_p < 0.01$ changes it by less than 1%) and emerges at $\theta = p_T/p \lesssim 1.7$ mrad. Charged particles below 5 mrad enter the far-forward system: the B0 spectrometer (5.5–20 mrad, charged tracks and photons above ~ 50 MeV), the Off-Momentum Detectors (≈ 45 –65% of beam rigidity in the far-forward acceptance table of Jentsch, Tu and Weiss; the routing code uses 40–60%), the Roman Pots at $z \approx 26/28$ m in the Yellow Report layout (32.5/34.3 m in epic-main; $\theta < 5$ mrad, ~ 60 –95% rigidity for off-rigidity tracks, AC-LGAD sensors with ~ 140 μm spatial and ~ 30 ps timing resolution), and the ZDC for neutrals [7,8,9]. A recoil at beam rigidity remains inside the beam envelope and is visible only in the Roman Pots, and only above the 10σ envelope cut; at low $|t|$ the intact recoil cannot be tagged at IP6 [10,51]. With an exponential coherent t -distribution the tag acceptance is analytic:

$$\varepsilon_{\text{tag}} = e^{-B p_{T,\text{cut}}^2} = 13.5\% \quad (p_{T,\text{cut}} = 0.20 \text{ GeV}, B = 50 \text{ GeV}^{-2}); \quad 4 \times 10^{-5} \quad (0.45 \text{ GeV})$$

Provenance of the cuts. The 0.20 GeV floor is the documented EIC requirement for high-acceptance (low-divergence) optics at top rigidity, corresponding to a beam divergence ≤ 73 μrad for 275 GeV protons [7,48], and matches the one documented light-ion number: coherent ${}^3\text{He}$ tagging limited to $p_T > 0.20$ GeV at IR-6 [10]. The 0.45 GeV value is our derivation (149 μrad high-divergence proton optics applied to the 300 GeV ${}^6\text{Li}$ momentum), not a documented number. No beam-optics parameters have been published for any lithium beam; if the ${}^6\text{Li}$ beam ran with proton-class divergence at this rigidity, the 10σ cut would fall at 0.5–0.7 GeV and eliminate the IP6 channel. This cut is the most consequential machine input to the measurement (assumption #1, §6).

The near-beam cut is angular; the reconstructed-level coherent sample. The 10σ envelope is a cut on the recoil *angle* (reconstruction-chain report §4), so the p_T threshold scales with the beam momentum: $p_T^{\text{cut}} = 10\sigma_\theta \cdot A \cdot p_u = 0.09 / 0.22 / 0.60$ GeV at 20.5 / 50 / 137.5 GeV/u for the proton-derived $\sigma_\theta = 73$ μrad , i.e. tag acceptances of 67% / 9% / 1.5×10^{-8} for $B = 50$ GeV^{-2} . The constant 0.20 GeV used for the figures of this report (13.5%) is close to the mid-energy value and far from the other two; the low-energy configuration is the natural home of the coherent program, and the lithium divergence itself remains undocumented (assumption #1). The ePIC pots surround a horizontal slot, so the cutout is a rectangle wide in x and tight in y . At the reconstructed level (money plot 6R: angular cut, slot-like cutout $|p_x| < 0.55$, $|p_y| < 0.22$ GeV, divergence smearing, two-azimuth spin-state-ratio fit) the mid-energy tagged sample is 2.7×10^6 events in year 1, the cutout's fake $\langle \cos 2\beta \rangle = +0.77$ is cancelled by the ratio, and the deformation coefficient of the first $|t|$ bin is recovered as $a_t(t_{\text{ref}}) = 0.117 \pm 0.007$ (1 yr) for 0.119 injected, with the flat term $a_e = 0.0101 \pm 0.0013$ for 0.010 — for a background-free, Z-identified coherent sample by assumption (the breakup budget is plans/06).

Coherence selection at IP6 and at IR-8. Standard EIC practice at IP6 is to establish coherence by vetoing breakup rather than by tagging the recoil: dissociation protons (rigidity 0.50) reach the Off-Momentum Detectors, neutrons the ZDC, de-excitation photons the B0 calorimeter and ZDC, and ${}^3\text{He}$ -like fragments (0.75) the Roman-Pot window [51]. Direct low- $|t|$ tagging is a central argument for the second interaction region, whose lattice adds a secondary beam focus ~ 45 m downstream where the beam size is $\sigma \approx 0.15 \times 0.1$ mm, allowing Roman Pots at $z = 44.0/45.5$ m to reach $p_T \approx 0$. The published intact-recoil efficiencies there are 47% (d), 32% (${}^3\text{He}$), 29% (${}^4\text{He}$), and 17.8% (${}^7\text{Li}$) at top energy, rising steeply at lower energies (${}^3\text{He}$ reaches 99.8% at 5×41) [10]; no ${}^6\text{Li}$ case has been published, and our interpolation is $\approx 20\%$. One degeneracy remains even there: a Roman-Pot hit measures only A/Z , so intact ${}^6\text{Li}$, α , and d at the same velocity are identical in trajectory and timing. The second-detector design addresses this with a proposed Z^2 -scaling Cherenkov counter behind the pots, currently at the R&D stage [8]. For the IP6 AC-LGADs no charge-discrimination study exists (signal amplitude is used only for position interpolation), and the minimum-angle ion acceptance there is not yet fully established [49]. This question, open item #19, determines whether IP6 coherent-tag purity is established event by event or statistically (§5).

4.2 • Rate model

$$\sigma_{\text{coh}} = f_{\text{coh}}(x) \sigma_{\text{DIS}}, \quad f_{\text{coh}}(x) = \frac{f_0}{1 + (x/x_{\text{coh}})^2}, \quad f_0 = 0.04 \text{ [0.02–0.08]}$$

No diffractive prediction exists for a light nucleus (the lightest published case is Ca), so the f_0 band brackets HERA ep diffraction (10–15% of DIS [44]) and heavy-nucleus saturation estimates (20–25% [45]). The scale $x_{\text{coh}} = 0.01$ marks where the coherence length $l_c = 0.105 \text{ fm}/x_p$ falls below the ${}^6\text{Li}$ size. The slope $B = 50 \text{ GeV}^{-2}$ (band 40–60) follows from $B = \langle r^2 \rangle / 3$ with the matter radius (the charge radius gives ≈ 57 [47]; gluonic distributions may be wider [42]); the exponential form is valid below the form-factor minimum at $|t| \approx 0.31 \text{ GeV}^2$ [11,12]. Parameterizing f_{coh} in Bjorken x rather than x_p is optimistic for high-mass diffraction, which the f_0 band is intended to cover. The resulting yields are 1.1×10^7 tagged events in year 1 and 1.1×10^8 over ten years, with best-super-bin $\delta A = 1.9 \times 10^{-3}$ and 6×10^{-4} respectively.

4.3 • The anchored amplitude

The $\cos 2\phi'$ modulation of the tagged yield carries two mechanisms. The *deformation* term is scaled from the polarized-deuteron CGC calculation of Mäntysaari, Salazar, Schenke, Shen, and Zhao [13], the only published calculation of polarization-dependent azimuthal coefficients in coherent diffraction on any polarized nucleus (all forward citations checked; the group's subsequent work is unpolarized [14,15]). Their coefficients, digitized in `coherent.MANTYSAARI_A2_DEUTERON`, show a_2 growing linearly in $|t|$ with $a_2(m=0) = -2a_2(m=\pm 1)$, arising from the 6% D-wave modulating the diffractive slope. For a fill with populations p_m the ensemble coefficient is proportional to P_{zz} :

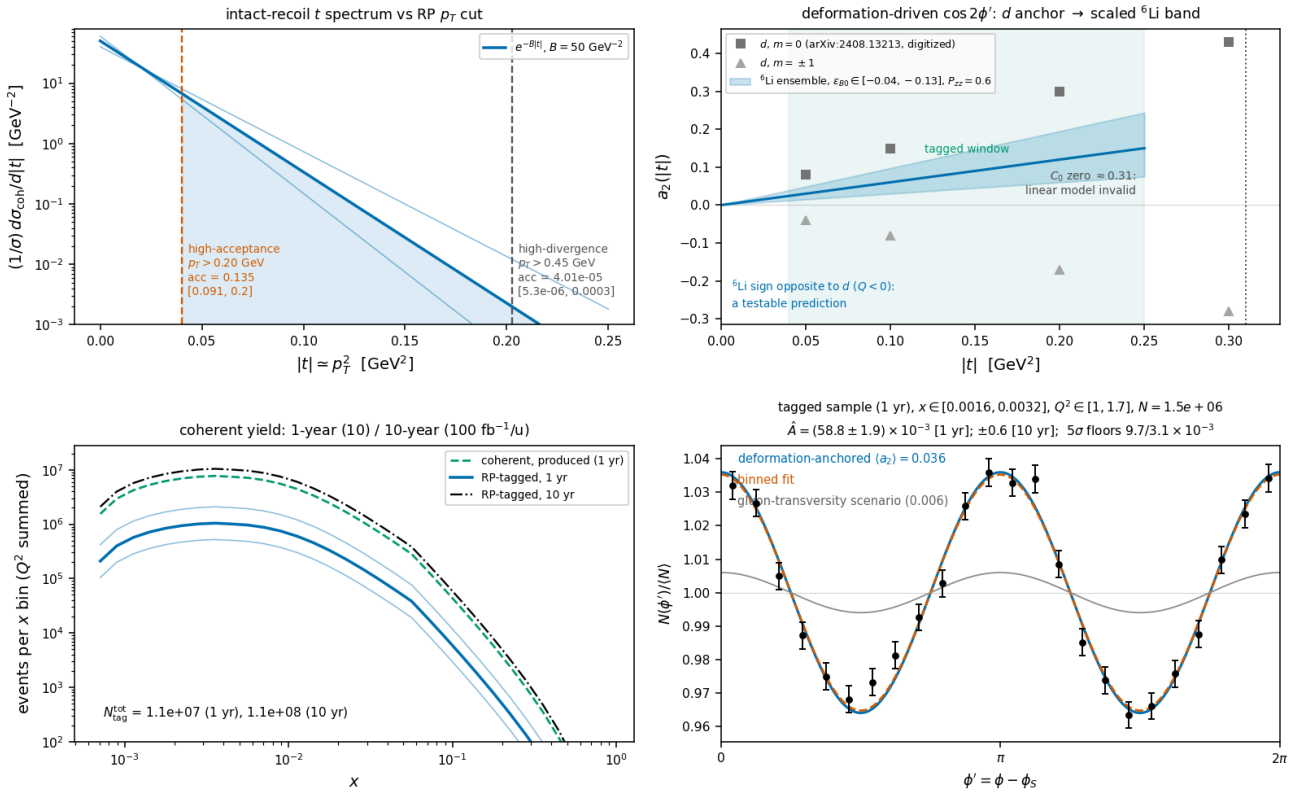
$$a_2(t; P_{zz}) = -\frac{P_{zz}}{4} \varepsilon_{B0} B |t|, \quad \langle a_2 \rangle_{\text{tag}} = a_2(p_{T,\text{cut}}^2 + 1/B) = 0.036 \text{ [0.018–0.059]}$$

Convention of the anchored coefficient. The anchor's Eq. (9) expands $d^2\sigma/d\Phi d|t| \propto 1 + 2\Sigma_n a_n e^{in\Phi}$, with Φ the azimuth of the produced vector meson — the momentum transfer, i.e. the recoil — relative to the polarization axis in the γ^*d frame. The $\cos 2\Phi$ modulation coefficient is therefore $2a_2$, and the $\langle a_2 \rangle$ quoted here and injected in money plot 6(d) understates the deformation modulation by a factor two (0.072 rather than 0.036 at $P_{zz} = 0.6$): this report's coherent numbers are conservative by $\times 2$ and are kept in the a_2 convention (the WP5 curves will carry the coefficient). The modulation is one of the *recoil* azimuth $\phi_t - \phi_S$, not of the electron azimuth used as the axis of panel (d); the reconstructed-level companion (reconstruction-chain report §7, money plot 6R) injects the coefficient $2a_2$ in the recoil azimuth and fits both azimuths.

$\varepsilon_{B0} \equiv \Delta B_0/B$ is scaled from the deuteron value of 0.21 using two indicators that disagree: the relative quadrupole deformation $Q/\langle r^2 \rangle$ of ${}^6\text{Li}$ is about one fifth of the deuteron's and opposite in sign, while the α -d asymptotic D/S ratio is disputed (a transfer measurement gives $\eta = +0.0003(9)$, consistent with zero [16], whereas older sub-Coulomb analyses favored $\eta \approx -0.01$). We adopt $\varepsilon_{B0} \in -(0.04\text{--}0.13)$ with default -0.08 , leaving near-zero deformation open; the 5σ floors cover its detection or exclusion. The width of the band reflects genuine theoretical uncertainty: even exact few-body calculations miss the ${}^6\text{Li}$ quadrupole cancellation by a factor of ≈ 3 [17], and the gluonic deformation need not share the cancellation observed in the charge sector. The sign flip relative to the deuteron is a testable prediction ($Q({}^6\text{Li}) < 0$). Tensor modulations of comparable size are established at these $|t|$: e-d $T_{20} = -(0.1\text{--}0.3)$ [19] and HERMES $A_{zz}(d) \approx -1\%$ at $x \approx 0.01$ [20], the latter attributed to coherent double scattering [21–23]. The flat gluon-transversity term is separately bounded at $3 \times 10^{-3}\text{--}10^{-2}$, from the bare lattice ϕ transversity [25] divided by the $\sim 10\times$ NPLQCD deuteron suppression [27] (both at unphysical pion masses), with the maximal Drell–Yan scenario [5] as the ceiling. The equal-rate population average behind $\langle a_2 \rangle$ carries a one-sided $\times 0.73$ rate-weighting systematic (audited; within the ε_{B0} band).

Separation of the two mechanisms. The mechanisms differ in shape: deformation is linear in t , sign-flipped relative to the deuteron, and expected to weaken toward small x_P (JIMWLK evolution washes out deformation [15]); exotic glue is flat in t and persists at small x , where it connects to the elliptic-gluon program [6]. In a two-component fit $a_2(t) = c_{\text{def}} \cdot |t| + a_g$ per x_P bin, both components exceed the statistical floor by $\geq 5\sigma$ at $100 \text{ fb}^{-1}/u$. Incoherent breakup is m-state-blind in this mechanism [13]: it dilutes the modulation but cannot mimic it.

Coherent $e^6\text{Li} \rightarrow e'X^6\text{Li(g.s.)}$, transversely tensor-polarized, $e(10) \times 6\text{Li}(50/u)$, $P_{zz} = 0.60$
 nario rates; modulation anchored on the polarized- d CGC calculation, PLB 858:139053 -- plans/06 SS6.4b; statistical only, backgrounds/purity and tensor RC unquanti



Money plot 6 — coherent channel, $e^6\text{Li} \rightarrow e'X^6\text{Li(g.s.)}$ (same beam, $P_{zz} = 0.60$). Top left: recoil t -spectrum vs both optics' p_T cuts with the slope band. Top right: digitized deuteron a_2 per m -state [13] vs the scaled ^6Li ensemble band (sign flip predicted); tagged window and form-factor-zero exclusion marked. Bottom left: tagged yield vs x for both programs with the f_0 band ($N_{\text{tag}} \approx 1.1 \times 10^7 / 1.1 \times 10^8$). Bottom right: ϕ' pseudo-data (1-year statistics) at the anchored $\langle a_2 \rangle = 0.036$ with the flat gluon-transversity scenario (gray); extraction $\hat{A} \pm 1.9 \times 10^{-3}$ (1 yr), $\pm 0.6 \times 10^{-3}$ (10 yr).

5 • Backgrounds for the coherent tag (full budget: plans/06)

BACKGROUND	WHY IT FAKES THE TAG	HANDLE
$\alpha + d$ breakup (threshold 1.474 MeV; the 2.186 MeV 3^+ state, $\Gamma = 24$ keV, decays $\sim 100\%$ $\alpha + d$ [39])	Both fragments at $R = 1.000$, same trajectory and velocity as intact ^6Li : tracking and ToF blind; only dE/dx ($\propto Z^2$: 9/4/1) separates	RP charge discrimination — open question #19 : no study exists for the IP6 AC-LGADs (amplitude is used for position; the documented Z-ID is the proposed IR-8 Cherenkov [8]). Fallback: two-component $ t $ fit (coherent $e^{-50 t }$ vs $\sim 10 \times$ flatter incoherent; crossover at 0.05–0.2 GeV^2 inside the window [41,42]); $e + \text{Pb}$ benchmark: 80–99% incoherent rejection [43]
Nucleon-emission breakup: $\alpha + p + n$ (3.70 MeV), $^3\text{He} + t$ (15.79 MeV) [39]	A fragment could pass unnoticed	$p \rightarrow \text{OMD}$ ($R = 0.50$), $n \rightarrow \text{ZDC}$, $^3\text{He} \rightarrow \text{RP window}$ ($R = 0.75$) [51]: vetoable; inefficiency is second-order (BeAGLE study to quantify)
$^6\text{Li}^*$ 3.5629 MeV (0^+ , $T = 1$, $\Gamma = 8.2$ eV) [39]	Particle-stable ($\alpha + d$ parity-forbidden, $\Gamma_\alpha \leq 6 \times 10^{-7}$ eV); γ -decays to a real $A = 6$, $Z = 3$ track	Isoscalar exchange cannot drive $T = 0 \rightarrow 1$; the boosted 0.1–0.4 GeV γ lands mostly in the B0 calorimeter (photons ≥ 50 MeV, 5.5–20 mrad [9]), tail in the ZDC. The 3.566 MeV $2^+ T = 1$ state always emits a nucleon. Every $T = 1$ state is γ - or nucleon-vetoable; only $T = 0$ states feed the blind $\alpha + d$ channel
Bethe–Heitler / QED Compton	Intact nucleus by construction	Calculable (t -dependence = charge FF^2 [11,12]); subtract and reuse as an RP-acceptance candle; the charge-quadrupole $\cos 2\phi$ is $< 10^{-3}$ for $ t < 0.15$ GeV^2 [12,17]
Beam-gas / halo; e' -side fakes	Accidental $R \approx 1$ tracks; $\pi /$ pair-symmetric e' at high y	Non-colliding bunches, ~ 30 ps timing, vertex association; standard DIS e' ID plus (e' , recoil) kinematic consistency

Concerning extraction systematics: in the spin-state ratio of §2, relative-luminosity errors enter only as a ϕ' -independent constant and any ϕ' -acceptance harmonic common to both spin states cancels bin by bin — a single-fill fit, by contrast, is biased by the acceptance's $\cos 2\phi'$ harmonic divided by P_{zz} ; the cancellation requires bunch-by-bunch alternation of the two states (or 10^{-4} stability of the harmonic between fills); the polarimetry scale $\delta P_{zz}/P_{zz} \approx 3\%$ is common with A_{zz} ; and radiative corrections for tensor observables have not been calculated, which is noted on every plot. At the level of interpretation, a $\Delta\Delta$ -isobar admixture can mimic Δ at leading twist [34] and conventional double scattering generates the low- x b_1 [21–23]; the Q^2 lever arm and the ${}^6\text{Li}/d$ comparison discriminate against both.

The ${}^6\text{Li}$ null test. Deformation-driven coherent modulations, the polarized-deuteron imaging signal [13], are bounded on ${}^6\text{Li}$ by its anomalously small quadrupole moment, $Q = -0.0806(6) \text{ fm}^2 \approx Q({}^7\text{Li})/50$ [35]. A sizable flat-in- $t \cos 2\phi$ on ${}^6\text{Li}$ therefore cannot be a shape effect; the mechanisms that are entangled for the deuteron separate in the ${}^6\text{Li}/d$ comparison.

6 • Critical assumptions

#	ASSUMPTION	IMPACT IF WRONG	REPLACEMENT PATH
1	RP p_T cuts: 0.20 GeV is the documented top-rigidity envelope scale (${}^3\text{He}$ precedent [10]); 0.45 GeV is our derivation; no Li optics exist — proton-class divergence would give 0.5–0.7 GeV	Tag acceptance anywhere from 13.5% to negligible at IP6; the physics case shifts to IR-8	Li beam-optics parameters from EIC C-AD; the IR-8 secondary focus removes the cut entirely [10]
2	Scenario coherent fraction $f_0 = 0.04$ [$\times 2/\div 2$], parameterized in Bjorken x	Coherent yields scale linearly; δA as $1/\sqrt{f_0}$	Real diffractive model — open question #18; concrete ask: an IP-Glasma α -d run [13]
3	Bag-moment normalization of Δ ported to per-nucleon ${}^6\text{Li}$ with 1/3 dilution; x-shape ours	Overall Δ scale; literature brackets both directions (NPLQCD suppression [27] vs binding enhancement [50])	Lattice moment for a light nucleus; theory engagement (plans/04)
4	ϵ_{B0} scaling from the deuteron (two dials disagreeing by $\times 3$)	Deformation amplitude anywhere in the band; sign flip robust	The same IP-Glasma α -d calculation; band quoted on every plot
5	Event-by-event Z-ID in the RPs is NOT assumed (purity via $ t $ -shape fit)	If Z-ID exists purity improves; if the incoherent slope model is off, the fit systematic grows	#19 answer from the ePIC far-forward group; BeAGLE incoherent spectra
6	Linear $a_2(t)$ valid only for $ t < 0.2 \text{ GeV}^2$; form-factor zero at 0.31 GeV^2 excluded, not modeled	Upper tagged window carries unmodeled structure	Form-factor-level model of the dip region (deuteron analog in [13])
7	TOY structure functions; no target-mass terms; $g_2 = g_2^{WW}$	Rates to $\times 1.5$; shapes are scenarios	PDF-grid backends exist; digitized theory curves scheduled (plans/02)
8	All luminosity in transverse-tensor states, $P_{zz} = 0.6$ in-ring placeholder (the two-state pattern $+0.6 / -1.2$ of the reconstruction-chain report shares the luminosity and is $1.5\times$ more precise)	N scales with the run-plan share; amplitude with P_{zz}	Run-plan bookkeeper supports splits; source demonstrations reach $\sim 95\%$ of maximal P_{zz} [37]
9	Impulse approximation; no FSI; tensor radiative corrections unknown	FSI distorts the recoil spectrum at larger $ t $; RC unquantified	Theory engagement (plans/04); the QED tail is calculable and doubles as calibration
10	Equal-rate population average for $\langle a_2 \rangle$; $t_{\min}$ neglected in the tag acceptance	One-sided -27% (rate weighting) and $\approx -10\%$ ($t_{\min}$) — inside the ϵ_{B0} / f_0 bands	Rate-weighted ensemble model (the IP-Glasma run supersedes both)

7 • Next steps

Near term: BeAGLE incoherent ${}^6\text{Li}$ spectra to quantify the veto efficiency and the $|t|$ -fit systematics (plans/02); VMC α -d/ α -t overlaps to replace the two-parameter radial forms (plans/04); the two-dimensional (α, β) pseudo-experiment closure,

done at the reconstructed level with an emulated Roman-Pot cutout (reconstruction-chain report §7, money plot 6R), to be repeated with the real acceptance maps (plans/03); and HepMC3 output with an ePIC chain test (plans/05). The inclusive projections have been repeated at the reconstructed level (money plots 5R and 7R, same report).

Theory: a rerun of the IP-Glasma polarized-deuteron setup [13] with an α -d cluster density, the highest-leverage input, which would replace both ϵ_{B0} indicators and the f_0 band at once; a lattice Δ moment for a light nucleus; the spin-3/2 structure-function basis for ^7Li ; tensor-observable radiative corrections; and the sum-rule connection between transversity-GPD moments and measured tensor form factors [30].

Machine and detector: lithium beam-optics parameters, which set the IP6 p_T cut (assumption #1); an answer to open item #19, whether the RP AC-LGAD amplitude can separate $Z = 3, 2, 1$ at identical rigidity and velocity; and the IR-8 secondary focus with its proposed Z^2 Cherenkov [8], which extends the tag toward $p_T \approx 0$ (^7Li : 17.8% published; ^6Li not yet studied [10]). The coherent- ^6Li program adds one more physics case for the second interaction region. Off-momentum halo determines how closely the pots approach the beam, and therefore the cut itself.

References

- [1] R. L. Jaffe, A. Manohar, *Nuclear gluonometry*, Phys. Lett. B 223 (1989) 218.
- [2] P. Hoodbhoy, R. L. Jaffe, A. Manohar, *Novel effects in deep inelastic scattering from spin-one hadrons*, Nucl. Phys. B 312 (1989) 571.
- [3] J. Maxwell et al., *Search for Exotic Gluonic States in the Nucleus*, JLab LOI12-16-006, arXiv:1803.11206.
- [4] E. Sather, C. Schmidt, *Size and scaling of the double-helicity-flip hadronic structure function*, Phys. Rev. D 42 (1990) 1424.
- [5] S. Kumano, Q.-T. Song, Phys. Rev. D 101 (2020) 054011, arXiv:1910.12523; Phys. Rev. D 101 (2020) 094013, arXiv:2003.06623.
- [6] Y. Hatta, B.-W. Xiao, F. Yuan, Phys. Rev. Lett. 116 (2016) 202301.
- [7] R. Abdul Khalek et al., *EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447, arXiv:2103.05419.
- [8] Far-forward/second-IR design: ATHENA collaboration, JINST 17 (2022) P10019, arXiv:2210.09048; *Precision Studies of QCD in the Low Energy Domain of the EIC*, arXiv:2211.15746 (§VIII); BNL LDRD 23-050 closeout (second-detector conceptual design, Mini-DIRC Z^2 Cherenkov), arXiv:2602.04636.
- [9] M. Pitt, PoS DIS2024 (2025) 259, arXiv:2409.02811; E.-C. Aschenauer et al., Phys. Rev. D 112 (2025) 036010, arXiv:2503.05908.
- [10] W. Chang, E.-C. Aschenauer, A. Jentsch, A. Kumar, Z. Tu, Z. Yin, *Opportunities for imaging light nuclei with a second interaction region at the EIC*, Phys. Rev. D 113 (2026) 032018, arXiv:2511.05638.
- [11] L. R. Suelzle, M. R. Yearian, H. Crannell, Phys. Rev. 162 (1967) 992.
- [12] G. C. Li, I. Sick, R. R. Whitney, M. R. Yearian, Nucl. Phys. A 162 (1971) 583.
- [13] H. Mäntysaari, F. Salazar, B. Schenke, C. Shen, W. Zhao, *Spatial imaging of polarized deuterons at the Electron-Ion Collider*, Phys. Lett. B 858 (2024) 139053, arXiv:2408.13213.
- [14] H. Mäntysaari, H. Roch, B. Schenke, C. Shen, W. Zhao, Phys. Rev. D 114 (2026) 014068, arXiv:2605.00454.
- [15] H. Mäntysaari, P. Singh, Eur. Phys. J. C 85 (2025) 1449, arXiv:2411.14934.
- [16] K. D. Veal, C. R. Brune, W. H. Geist et al., Phys. Rev. C 60 (1999) 064003.
- [17] R. B. Wiringa, R. Schiavilla, Phys. Rev. Lett. 81 (1998) 4317, nucl-th/9807037.
- [19] M. Bouwheuis et al. (NIKHEF), Phys. Rev. Lett. 82 (1999) 3755; D. Abbott et al. (JLab t20), Phys. Rev. Lett. 84 (2000) 5053; D. M. Nikolenko et al. (VEPP-3), Phys. Rev. Lett. 90 (2003) 072501.
- [20] A. Airapetian et al. (HERMES), Phys. Rev. Lett. 95 (2005) 242001, hep-ex/0506018.
- [21] N. N. Nikolaev, W. Schäfer, Phys. Lett. B 398 (1997) 245.
- [22] K. Bora, R. L. Jaffe, Phys. Rev. D 57 (1998) 6906, hep-ph/9711323.
- [23] J. Edelmann, G. Piller, W. Weise, Phys. Rev. C 57 (1998) 3392.
- [25] W. Detmold, P. E. Shanahan, Phys. Rev. D 94 (2016) 014507 [Erratum: Phys. Rev. D 95 (2017) 079902], arXiv:1606.04505.
- [27] F. Winter et al. (NPLQCD), Phys. Rev. D 96 (2017) 094512, arXiv:1709.00395.
- [30] W. Cosyn, B. Pire, Phys. Rev. D 98 (2018) 074020, arXiv:1806.01177; W. Cosyn et al., Phys. Rev. D 99 (2019) 094035.
- [34] M. Nzar, P. Hoodbhoy, *Estimation of the double helicity flip deuteron structure function*, Phys. Rev. D 45 (1992) 2264.
- [35] I. Angeli, K. P. Marinova, At. Data Nucl. Data Tables 99 (2013) 69; N. J. Stone, At. Data Nucl. Data Tables 111-112 (2016) 1.
- [37] M. M. Dalton, A. Deur, C. D. Keith, Eur. Phys. J. A 61 (2025) 111, arXiv:2504.21177.
- [39] D. R. Tilley et al., *Energy levels of light nuclei $A = 5, 6, 7$* , Nucl. Phys. A 708 (2002) 3 (TUNL).
- [41] LEPS collaboration, coherent ϕ on ^4He , arXiv:1711.01095.
- [42] STAR collaboration, Phys. Rev. Lett. 128 (2022) 122303, arXiv:2109.07625.
- [43] W. Chang et al., Phys. Rev. D 104 (2021) 114030, arXiv:2108.01694.
- [44] H1 collaboration, Eur. Phys. J. C 72 (2012) 2074, arXiv:1203.4495.
- [45] T. Lappi, Nucl. Phys. A 830 (2009) 403c, arXiv:0907.4588; H. Kowalski, T. Lappi, C. Marquet, R. Venugopalan, Phys. Rev. C 78 (2008) 045201, arXiv:0805.4071.

[47] I. Sick, arXiv:1505.06924.

[48] ECCE consortium, exclusive/diffractive/tagging studies, NIM A 1052 (2023) 168238, arXiv:2208.14575.

[49] J. R. Pybus, S. Li, L. Calero-Diaz, *Polarized nuclear DVCS at the EIC* (e+³He), arXiv:2606.11491.

[50] G. Atoian et al. (EPIOS), *Realizing the scientific program with polarized ion beams at EIC*, Phys. Rev. C 113 (2026) 060501, arXiv:2510.10794.

[51] IR-8 far-forward tagging of nuclear breakup, arXiv:2501.12410 (bibliographic details to be completed; not in the local reference index).

Appendix A • Modification history

The main text states the current content only; this appendix records what changed and when (the repository history carries the full diffs).

DATE	REVISION
2026-08-10	First version (development run 5): money plots 5 and 6 with the moment-constrained $\Delta(x,Q^2)$ and separate 1-year / 10-year projections, the physics note with a verified bibliography (docs/note_cos2phi_coherent_6Li.md); same day, first audit round — fact-check fixes to formulas, port fidelity and documents.
2026-08-11	Second audit round (formulas against the literature, implementation, references, extraction logic); money plot 7 (the extracted Δ as structure-function data points) added.
2026-08-17	Dedicated fact checks of the Sather–Schmidt moment usage (a Δ^{++} bag calculation without x-shape) and of the far-forward detection chain; phase-space figure with the analysis binning; display mathematics typeset at build time; academic-register line edit.
2026-08-24	§4.3 convention of the anchored coefficient: the anchor's Eq. (9) expands $1 + 2\sum a_n e^{in\Phi}$, so the $\cos 2\Phi$ coefficient is $2a_2$ and this report's coherent numbers are conservative by $\times 2$; the reconstruction-chain report created as the reconstructed-level companion.
2026-08-25	Estimator: the spin-state-sorted ratio of $m = \pm 1$ -rich and $m = 0$ -rich bunches replaces the single-state fit as the estimator of the measurement ($0.67\times$ error; bunch-by-bunch alternation required), with the reconstructed-level results 5R/7R quoted; the near-beam cut stated as angular ($0.09 / 0.22 / 0.60$ GeV $\rightarrow$ 67% / 9% / 10^{-8} at 20.5 / 50 / 137.5 GeV/u) with the slot-like cutout and the 6R numbers; the sign convention of the b_1, b_2 terms stated (HJM/HERMES/Cosyn: $A_{zz} = -(2/3)b_1/F_1$); the inversion corrected to $\Delta = -\hat{A} \gamma^2 D_\phi / (1-\gamma)$ (also on money plot 7); the sweet-spot γ^2 statement corrected ($x \leq 0.14, \gamma^2 \leq 0.023$) with the $O(\gamma^2)$ b_1 leakage; "self-normalized within a single fill" replaced by the ratio statement in §5; OMD window 45–65%, Roman-Pot z 26/28 m (Yellow Report) versus 32.5/34.3 m (epic-main), 149 μ rad; caveats where reconstructed-level numbers are quoted (K model dependence, hadronic- γ regime, background-free 6R); reference [51] flagged incomplete; test count 94 + 24. Figures and projections unchanged.
2026-08-25	Modification history moved to this appendix; the main text restated as current content.

Generated from the PolarizedLithiumSim repository: `evgen/scripts/money_cos2phi.py` , `evgen/scripts/money_cos2phi_coherent.py` , `evgen/scripts/money_delta_extraction.py` , `evgen/scripts/phase_space_bins.py` ; models in `polli_fastsim/delta_models.py` and `evgen/polligen/coherent.py` ; assembled by `reports/build_report.py` (display math typeset at build time, matplotlib mathtext/STIX). 94 + 24 tests pass. All rates use TOY/scenario inputs — bands, not predictions. External numbers verified against primary sources (nuclear data: TUNL; detector: EIC YR, ATHENA, ePIC and second-IR documents; theory: original papers), including dedicated fact checks of the Sather–Schmidt moment usage and the far-forward detection chain (Appendix A). Reference numbering is shared across revisions; retired entries are omitted. Target venue: INT program on polarized ion beams at the EIC, March 22 – April 2, 2027.